Overcoming Visual Clutter in Vision Language Action Models via Concept-Gated Visual Distillation

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***Abstract*— Vision-Language-Action (VLA) models demonstrate impressive zero-shot generalization but degrade sharply in cluttered scenes, a failure we term the "Precision-Reasoning Gap" and hypothesize to reflect background-induced feature dilution corrupting the visual grounding required for precise manipulation. We propose Concept-Gated Visual Distillation (CGVD), a training-free, model-agnostic inference-time framework that parses the instruction into a safe set, pairs a supplied distractor vocabulary with a two-layer target refinement that penalizes false positives, and inpaints the gated distractor regions, preserving scene context outside the edits. Compared against the un-augmented π_0 and GR00T policies in SimplerEnv, CGVD substantially improves robustness to semantic clutter—on *put spoon on towel* with 18 semantic distractors, π_0's success rate rises from 43.0% to 77.5% (Table III)—and, in one exploratory 5-seed condition with compositional prompts, improves attribute adherence. A vocabulary-matched BYOVLA reimplementation reaches 32.5% at the same condition, at roughly thirty times CGVD's per-chunk cost, in a dense-clutter regime outside its design envelope (§IV-E). The protection is task- and distractor-type-conditional: on *put carrot on plate*, where contextual clutter aids the baseline policies, CGVD provides no benefit at any tested density and often significantly underperforms them. Runtime compositing adds roughly 16 ms of per-frame overhead after a sub-second per-episode initialization. These results support inference-time visual distillation as a practical defense against semantically confusable clutter for frozen VLA policies in static scenes with a known clutter vocabulary.**

I. INTRODUCTION

Vision-Language-Action (VLA) models [1]–[5] ground large language models into robotic control, enabling robots to follow open-vocabulary instructions like "Put spoon on towel" without task-specific training. These models promise a future where robots can operate in unstructured, human-centric environments.

However, a significant gap remains between these models' semantic reasoning and their geometric precision in deployment conditions. While VLAs excel in curated, clutter-free environments, their performance degrades precipitously in the presence of visual clutter (Fig. 1) [6]. We term this phenomenon the Precision-Reasoning Gap: the model successfully identifies the target object conceptually, yet fails to manipulate it precisely. We hypothesize that attention corruption from surrounding distractors dilutes the latent representation used for spatial planning; under this hypothesis, feature dilution would manifest as high-variance trajectories and manipulation failure. Our experiments (§IV) measure the end-to-end effect of removing distractors without directly instrumenting attention, so we present this account as a working hypothesis rather than an established mechanism.

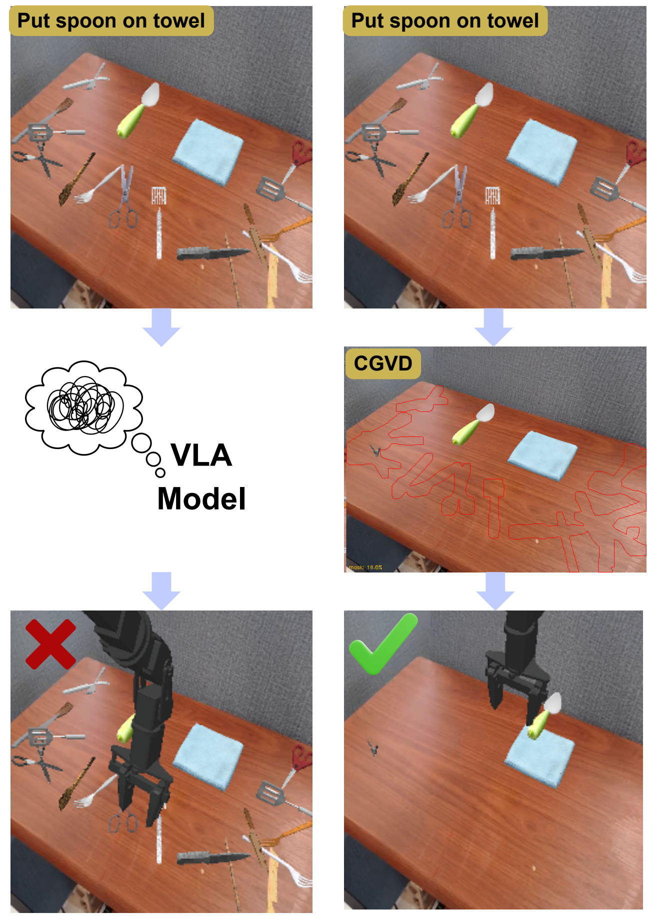

Fig. 1. Manipulation under clutter: a standard VLA (left) suffers object confusion, while CGVD (right) places the target spoon on the towel.

This degradation is not uniform across distractor types: failure concentrates around distractors sharing visual or semantic properties with the target, consistent with prior reports [6]. While VLAs may exhibit resilience to arbitrary clutter via large-scale pre-training, they remain brittle to semantically confusable objects; for example, a fork scattered near a target spoon triggers conflicting visual tokens within the same affordance category, causing the policy to attend to or even grasp the wrong object.

Existing mitigations fall into three categories (§II): observation-grounding methods such as OBEYED-VLA [7] augment the VLA with a trained perception module producing object-centric, geometry-grounded observations and fine-tune the policy on clean demonstrations; inference-time interventions such as BYOVLA [8] identify distractors with

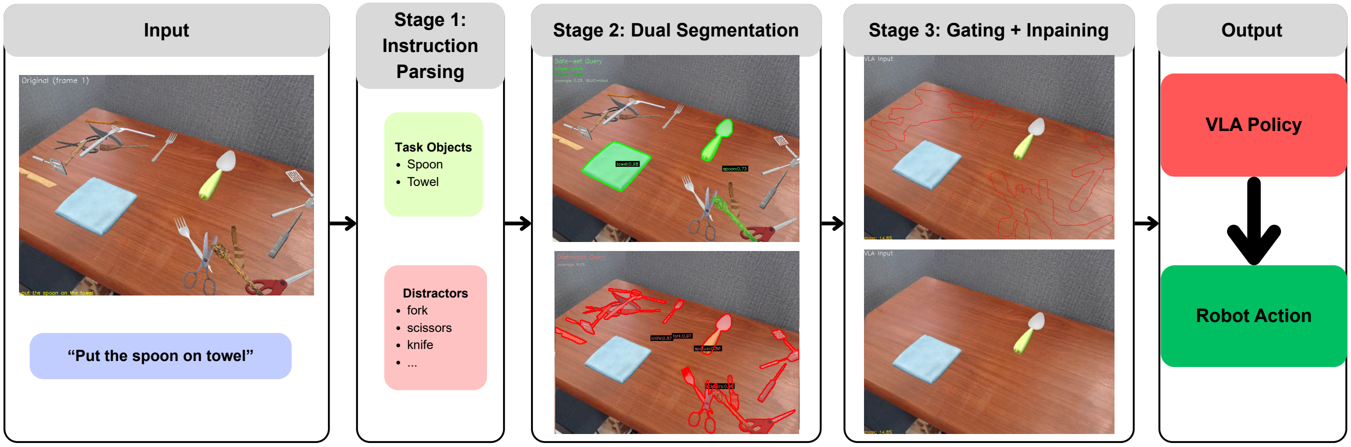

Fig. 2. The CGVD pipeline. **Stage 1:** the instruction is parsed into a safe set (target, anchor); the distractor vocabulary is supplied. **Stage 2:** SAM 3 segments both sets independently. **Stage 3:** set-theoretic gating subtracts the safe-set mask from the distractor mask; LaMa inpaints the result into the distilled observation passed to the VLA.

a VLM (a GPT-4o dependency) and probe per-region sensitivity through repeated VLA forward passes under threshold-based protection; and training-time augmentation [9]–[11] generates cluttered training data at the cost of retraining, without deployment-time guarantees.

To bridge this gap without retraining, we propose Concept-Gated Visual Distillation (CGVD): a model-agnostic inference framework leveraging vision foundation models [12] to restructure observations before they reach the policy. CGVD parses the instruction into target and anchor, segments distractors and task-relevant entities independently, and uses set-theoretic subtraction to exclude the target from the distractor mask *conditional on correct safe-set segmentation*: no pixel segmented as safe is inpainted, but an object the segmenter misses is unprotected (§IV-D); the gated distractors are then inpainted [13], preserving surrounding context.

Our contributions are as follows:

- **Concept-Gated Visual Distillation (CGVD):** a training-free, model-agnostic inference framework removing distractors from VLA observations via language-grounded segmentation and inpainting while preserving scene context.
- **Interaction-Aware Masking Logic:** a set-theoretic cross-validation pipeline that overcomes the semantic confusion of open-set segmenters (which score prompts independently), penalizing false positives and spatially disambiguating true targets from confusable distractors.
- **Systematic clutter-density evaluation:** We evaluate CGVD on two VLAs (π_0, GR00T) within the SimplerEnv benchmark over more than 20,000 episodes, comparing each policy with and without CGVD under matched seeds: CGVD prevents policy collapse under dense *semantic* clutter—77.5% against the unaugmented 43.0% (π_0, *put spoon on towel*, 18 semantic distractors; Table III)—improves attribute adherence in an exploratory compositional-prompt study, and *underperforms* the baseline where contextual clutter aids the policy (§IV-B).

II. Related Work

A. Vision-Language-Action Models

VLA models such as RT-2 [2], OpenVLA [1], and Octo [3] leverage internet-scale pre-training for zero-shot generalization, yet struggle with high-precision geometric grounding in clutter [6]. We hypothesize that the static patch resolution of ViT backbones makes such architectures susceptible to feature dilution [6]: background clutter competing with task-relevant signals for representational capacity.

B. Vision Foundation Models for Robotic Perception

Vision Foundation Models such as SAM 3 [12] localize objects from free-form text; robotics has mostly used them to *add* information, for instance via visual prompting [14]. Our work inverts this paradigm, leveraging the open-set discriminative power of VFMs to suppress task-irrelevant regions instead; the resulting mask is an information bottleneck only in an informal, analogical sense [15], with no learned rate–distortion trade-off.

C. Robustness via Training-Time Augmentation

Domain randomization and large-scale pre-training [16] have not made VLAs robust to cluttered-scene shifts [17]. Training-time augmentation generates cluttered or counterfactual data: ROSIE [9], GenAug [10], and NICE [11], whose pixel-space *scene surgery* is the closest prior to our core operation. CGVD performs a related pixel-space edit at *inference* time on a frozen policy, trading retraining for run-time perception risk.

D. Inference-Time Attention Intervention

Distracting Token Pruning (DTP) [18] suppresses text-misaligned visual tokens in feature space; we conjecture such pruning struggles once distractor and target tokens are entangled by early self-attention, and instead intervene in pixel space, before tokenization. Unlike Visual Prompting [14], which adds information, we remove it. Two prior methods also operate on the observation: OBEYED-VLA

[7], whose trained perception module and policy fine-tuning contrast with CGVD's training-free design, and BYOVLA [8], whose run-time intervention §IV-E measures directly.

III. Methodology

CGVD is a training-free, inference-time perception wrapper around any VLA policy (Fig. 2): given instruction l and observation $o_t \in \mathbb{R}^{H\times W\times 3}$, a distillation function ϕ produces $\hat{o}_t = \phi(o_t, l)$ with distractors replaced by background, and the policy acts on $a_t = \pi(\hat{o}_t, l)$. Since ϕ touches only the observation interface, it is architecture-agnostic; the key insight is that the instruction already specifies which objects must remain visible, while the clutter vocabulary is supplied separately (§III-A).

A. Concept-Gated Decomposition

Parsing the instruction l yields a target concept c_{tgt} and an anchor c_{anc} ("put spoon on towel" gives `spoon` and `towel`), defining two sets:

- The **safe set** $\mathscr{S} = \{c_{\text{tgt}}, c_{\text{anc}}\}$: entities that must remain visible. The robot arm is also always preserved, via a separate proprioceptive channel (§III-E–III-F) rather than the safe-set mask.
- The **distractor set** $\mathscr{D} = \{d_1, \ldots, d_K\}$: semantic categories to be treated as clutter (e.g., spatula, fork, knife).

The safe set is derived deterministically from the instruction; the distractor set, by contrast, is an externally supplied vocabulary. In our experiments, $\mathscr{D}$ is re-derived at every episode reset from the category labels of the actually spawned objects (§IV-A), so CGVD receives the exact ground-truth clutter inventory per episode; at deployment it would have to come from the integrator as a site vocabulary or from an external model. Where prior work uses an LLM to determine relevance [8], our parsing is deterministic and needs no API in this closed-vocabulary setting, a different point on a coverage-versus-cost trade-off; §IV-D measures the dependence directly.

B. Text-Prompted Instance Segmentation

Both the safe set and distractor set are segmented from the observation using SAM 3 [12]. The two sets produce independent mask channels:

$$M_{\text{dist}} = \bigcup_{d_k \in \mathscr{D}} \textsc{Seg}(o_t, d_k), \tag{1}$$

$$M_{\text{safe}} = \textsc{Seg}(o_t, c_{\text{tgt}}) \cup \textsc{Seg}(o_t, c_{\text{anc}}), \tag{2}$$

where $\textsc{Seg}(o_t, c)$ is the union of all instance masks SAM 3 returns for concept c. As a computational optimization, the vision encoder is executed only once, on the initialization frame, with its masks reused across the episode.

C. Two-Layer Target Refinement

Open-set segmenters evaluate text prompts independently, so single-pass detection suffers semantic confusion: a spatula can yield high confidence for "spoon." A two-layer refinement on the initial frame ensures the true target survives distillation.

Layer 1: Cross-Validation. Each target instance s_i receives a genuineness score—the confidence differential between its safe-set and distractor identities:

$$g(s_i) = \sigma_{\text{safe}}(s_i) - \max_{\substack{d_j \in \mathscr{D} \\ \text{IoU}(s_i, d_j) > \eta}} \sigma_{\text{dist}}(d_j), \tag{3}$$

where σ_{safe} and σ_{dist} are object confidences, and η is an IoU threshold ($\eta = 0.30$). When no distractor satisfies $\text{IoU}(s_i, d_j) > \eta$, the common case for an isolated target, the maximum is defined as 0 and $g(s_i)$ reduces to $\sigma_{\text{safe}}(s_i)$. Genuine targets yield $g > 0$, imposters $g < 0$; negative values are preserved to penalize false positives in the next layer.

Layer 2: Spatial Disambiguation. Since the target mask may still contain fragments or multiple disjoint objects, each connected component C_k is scored:

$$\text{score}(C_k) = \big(1 + g^*(C_k)\big) \cdot \sigma^*(C_k). \tag{4}$$

Here, $g^*(C_k)$ is maximum genuineness and $\sigma^*(C_k)$ is peak safe-set confidence. Only the top-scoring component is retained, so the pipeline assumes a single target instance per scene (§V).

D. Concept-Gated Mask Composition

The refined masks are combined into a final inpainting mask via set-theoretic gating:

$$M_{\text{inp}} = \text{dilate}(M_{\text{dist}}, r_d) \setminus \text{dilate}(M_{\text{safe}}, r_s), \tag{5}$$

where r_d is a distractor dilation radius, and $r_s \geq r_d$ is a safe-set dilation radius that creates a protective buffer; the implementation enforces $r_s \geq r_d$ by construction, and both are 11 px in our configuration. Masks are binarized at 0.5 before dilation to eliminate soft-value artifacts.

E. Clean Scene Generation via Inpainting

We generate a single clean scene, the workspace with distractors removed, by applying LaMa [13] to the initial frame with the mask:

$$M_{\text{lama}} = M_{\text{inp}} \cup \text{dilate}(M_{\text{robot}}, r_e), \tag{6}$$

where M_{robot} is the robot mask (§III-F) and r_e a robot dilation radius (20 px). Inpainting the robot region synthesizes the background *behind* the arm, leaving no stale robot pixels once it moves. LaMa fills the masked regions with photorealistic background texture; the clean scene is computed once per episode and cached. A consequence is that content occluded by the arm at $t{=}0$ is synthesized rather than observed for the rest of the episode (§V).

F. Temporally Consistent Compositing

At each timestep $t > 0$, the distilled observation is produced by alpha-blending the live camera frame with the cached clean scene:

$$\hat{o}_t = \alpha \odot \hat{o}_{\text{clean}} + (1 - \alpha) \odot o_t, \tag{7}$$

where the compositing mask $\alpha \in [0,1]^{H\times W}$ is computed *once* at $t{=}0$ and held fixed: the binarized, safe-set-gated distractor mask is feathered with a Gaussian blur ($\sigma = 3.0$), clamped to 1 in core distractor regions and 0 in the (dilated) target region. So that inpainting artifacts never obscure the arm, robot pixels are overwritten onto the composite every step—the only per-frame operation. In simulation this robot mask comes from SimplerEnv's renderer segmentation buffer (per-link actor IDs): pixel-perfect, jitter-free, and *simulation-privileged*; §IV-D measures the deployable substitutes.

Two properties of this design should be stated explicitly. First, $\hat{o}_t$ is the only visual input the policy receives. Second, the characteristic failure mode of observation editing is *false-positive erasure*: an object misclassified as a distractor—or entering the workspace after $t{=}0$, where the fixed α can blend it toward the cached background—is removed from the policy's view while remaining in the world, and the arm can collide with what it cannot see. Our evaluation is therefore scoped to static workspaces with no human entry, a fixed camera, and a known clutter vocabulary.

IV. Experiments

We evaluate CGVD across two VLA architectures on tabletop manipulation tasks with controlled distractor injection, asking whether it improves clutter robustness across architectures and tasks, how performance scales with distractor density, and how its components contribute at the operating point tested.

A. Experimental Setup

Environment. We evaluate in SimplerEnv [19], whose real-to-sim correlation supports *policy-level* transfer. It does not bound CGVD's perception-level risk: SAM 3 [12] and LaMa [13] here process rendered frames, itself a domain shift, and two inputs are simulation-privileged (the renderer robot mask, §III-F; the vocabulary, §III-A). Conclusions are scoped to simulation (§V).

All experiments use the WidowX arm with a single fixed third-person camera, matching the Bridge training setup, on two Bridge tasks: *put spoon on towel* and *put carrot on plate*.

Policies. π_0 denotes the open-source `open-pi-zero` reproduction of π_0 [5]; GR00T denotes `GR00T-N1.6-bridge` [4].[1] Both are frozen; CGVD touches only their image input.

Distractors. We introduce a three-way distractor taxonomy: *Semantic* (high semantic proximity and shared visual characteristics with the target), *Random* (no similarity), and *Attribute* (same category and form, different physical properties); [6] documents the magnitude of clutter-induced degradation with a continuous clutter measure rather than these types. Assets come from RoboCasa [17] and YCB [20], placed collision-aware on a fixed 6 cm grid (one object per cell); an unplaceable object falls back to a collision-free random position or is hidden off-scene—episodes are never dropped. In 1,400 audited re-executions of the headline condition (identical placement generator and seeds), 18.7% of episodes realized fewer than the nominal 18 objects (mean 17.6), making realized-density trends, if anything, conservative. The vocabulary $\mathscr{D}$ is re-derived per episode from the actually-spawned assets' category labels (`rc_fork_0` $\rightarrow$ "fork"): it coincides exactly with each episode's injected inventory (§III-A).

Protocol. We report success rate (SR) via SimplerEnv's built-in per-task predicate over 60-step episodes: 20 episodes $\times$ 10 seeds per condition (200 episodes; Table II's attribute study uses 5 seeds), with matched seeds between arms. Seed $s \in \{0,\ldots,9\}$ seeds the RNGs; episode e uses canonical initial layout $(s{+}e)$ mod 24 and distractor seed $10^4 s + e$, so both arms see identical states and clutter. Counts are sampled at $\{0,4,8,10,14,18\}$; no episodes were excluded. We compare CGVD against the un-augmented policies throughout, and against a reimplementation of BYOVLA [8] at the headline condition (§IV-E). OBEYED-VLA's release covers only its perception module, so an end-to-end comparison was infeasible; DTP's [18] only advertised release, an anonymized review-time repository, has expired.

Analysis. We report episode-level mean success rates with the SD of the 10 per-seed rates as dispersion; differences carry 95% CIs from the per-seed paired deltas (t_9). Clustering by seed is negligible (one-way ICC across all arms: median -0.02, max 0.06). The headline contrast—π_0, *put spoon on towel*, 18 semantic distractors—is the single confirmatory comparison ($\Delta = +34.5$ pp, 95% CI $[+20.8, +48.2]$, paired t $p = 3{\times}10^{-4}$); all other contrasts are exploratory, and deltas whose CI contains zero are within run-to-run noise. The 0-distractor cells are shared between the semantic and random conditions (physically identical), and the one condition executed twice end-to-end differed between batches by at most 3.0 pp, a direct replication estimate.

Implementation. All CGVD parameters, held fixed across every reported experiment: SAM 3 checkpoint `facebook/sam3`; LaMa big-lama (`simple-lama-inpainting`); confidence thresholds 0.30 (safe set) and 0.20 (distractors); $\eta = 0.30$; $r_d = r_s = 11$ px; $r_e = 20$ px; compositing $\sigma = 3.0$; mask binarization 0.5; minimum connected component 50 px; one warm-up frame; clean scene cached once per episode; prompts are the instruction's noun phrases and $\mathscr{D}$'s category labels. Values were set on small pilot rollouts and frozen before all reported runs (verified from version-control history). Code, configuration, and evaluation scripts will be publicly released upon acceptance.

B. Main Results

Table I reports the full grid (both policies, tasks, and clutter types at 0–18 distractors) with per-seed dispersion and paired CIs.

In the presence of semantically confusing clutter, baseline performance degrades precipitously and CGVD prevents this collapse for both policies; the gap also widens with clutter, at a per-seed slope of $+1.54$ pp per distractor for π_0 ($p{<}0.001$) and $+0.74$ for GR00T ($p{=}0.02$). A similar, smaller benefit appears under random clutter at high density.

[1] Checkpoints: `bridge_beta` from `github.com/allenzren/open-pi-zero`; `nvidia/GR00T-N1.6-bridge`. Ref. [4] documents GR00T N1; N1.6 is a revision with a different backbone and action head.

TABLE I

MAIN RESULTS (NUMERIC). SUCCESS RATE (%), MEAN ± SD OVER 10 MATCHED SEEDS (200 EPISODES/CELL); Δ WITH 95% CI FROM PER-SEED PAIRED DELTAS (t_9), BOLD WHEN THE CI EXCLUDES ZERO.

		Semantic distractors						Random distractors					
		π_0			GR00T			π_0			GR00T		
Task	**#D**	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]
spoon on towel	0	82.0	85.0	+3.0 [-5.6,+11.6]	49.0	51.0	+2.0 [-6.1,+10.1]	82.0	85.0	+3.0 [-5.6,+11.6]	49.0	51.0	+2.0 [-6.1,+10.1]
	4	59.0	73.0	**+14.0 [+5.4,+22.6]**	47.5	46.0	-1.5 [-8.7,+5.7]	70.5	78.0	+7.5 [-0.8,+15.8]	48.0	48.0	+0.0 [-7.0,+7.0]
	8	51.0	73.5	**+22.5 [+10.7,+34.3]**	28.5	41.0	**+12.5 [+6.4,+18.6]**	62.5	74.0	**+11.5 [+1.7,+21.3]**	44.0	43.5	-0.5 [-12.5,+11.5]
	10	49.0	72.5	**+23.5 [+12.8,+34.2]**	29.5	43.0	**+13.5 [+6.1,+20.9]**	60.0	78.5	**+18.5 [+10.2,+26.8]**	41.0	47.0	+6.0 [-3.0,+15.0]
	14	52.5	75.5	**+23.0 [+14.7,+31.3]**	30.0	41.0	**+11.0 [+3.3,+18.7]**	64.5	77.5	**+13.0 [+1.7,+24.3]**	33.0	42.0	+9.0 [-0.9,+18.9]
	18	43.0	77.5	**+34.5 [+20.8,+48.2]**	31.0	43.5	**+12.5 [+4.0,+21.0]**	57.5	70.5	**+13.0 [+2.0,+24.0]**	28.0	45.0	**+17.0 [+7.6,+26.4]**
carrot on plate	0	58.5	54.5	-4.0 [-11.7,+3.7]	32.0	28.0	-4.0 [-12.2,+4.2]	58.5	54.5	-4.0 [-11.7,+3.7]	32.0	28.0	-4.0 [-12.1,+4.1]
	4	57.5	52.5	-5.0 [-11.5,+1.5]	41.5	32.0	**-9.5 [-12.1,-6.9]**	56.0	53.0	-3.0 [-10.8,+4.8]	36.5	30.0	-6.5 [-16.5,+3.5]
	8	62.0	53.0	-9.0 [-21.1,+3.1]	42.5	31.0	**-11.5 [-17.1,-5.9]**	55.0	52.5	-2.5 [-10.8,+5.8]	43.0	28.5	**-14.5 [-21.1,-7.9]**
	10	61.5	53.5	**-8.0 [-15.0,-1.0]**	42.5	29.0	**-13.5 [-20.7,-6.3]**	56.0	55.0	-1.0 [-10.7,+8.7]	40.0	28.5	**-11.5 [-18.7,-4.3]**
	14	58.5	51.5	-7.0 [-17.4,+3.4]	32.0	27.5	-4.5 [-9.7,+0.7]	53.5	52.0	-1.5 [-13.8,+10.8]	39.0	37.5	-1.5 [-9.2,+6.2]
	18	56.0	47.0	**-9.0 [-17.7,-0.3]**	32.0	28.5	-3.5 [-10.0,+3.0]	49.5	54.0	+4.5 [-6.9,+15.9]	38.0	41.0	+3.0 [-3.6,+9.6]

Conversely, on *put carrot on plate* the baseline policies improve slightly at moderate densities before degrading, likely because moderate contextual clutter better aligns with the object-rich scenes of their pre-training distributions, providing visual anchors for reasoning.

Here CGVD never significantly outperforms the baseline, and at several densities it significantly underperforms it, most sharply for GR00T (Table I). This suggests that when a task naturally benefits from contextual clutter, masking deprives the VLA of useful environmental reasoning, and inpainting artifacts can further disrupt spatial geometry; the intervention is therefore task- and distractor-type-conditional rather than universally beneficial.

Qualitative attention visualizations consistent with this working hypothesis are provided with the released code (final-query-token attention over the 256 SigLIP tokens, averaged across the PaliGemma backbone). Such visualizations are entailed by the intervention itself, since removing a distractor's pixels removes attention on them, and no attention statistic is measured in this paper; alternatives—restoring the observation toward the clutter-sparse Bridge training distribution, or simply deleting the wrong-grasp option—remain consistent with our data. Indeed, the mean-color-fill ablation in §IV-D indicates that fill appearance, rather than distractor semantics alone, carries a large share of the effect.

C. Fine-Grained Semantic Grounding

The precision-reasoning gap is most acute when distractors share the target's semantic class but differ in attributes: complex queries like "Put spoon with green handle on towel" are often reduced to a bag-of-words, entangling the target with fully green objects or dropping modifiers. We therefore evaluate a green-handled spoon target among same-category spoons of other colors or materials (the *Attribute* type, §IV-A), across 0–4 distractors and two prompt structures, Simple ("Put green spoon on towel") and Complex ("Put spoon with green handle on towel"); CGVD receives the same instruction, with its attribute phrase as the SAM 3 target prompt. This study covers only π_0, one task, and 0–4 distractors, a narrower slice than Table I, so attribute claims are scoped accordingly.

TABLE II

ATTRIBUTE DISTRACTOR SENSITIVITY ON *put spoon on towel* (π_0, 100 EPISODES/CELL, 5 MATCHED SEEDS). Δ: 95% CI FROM PER-SEED PAIRED DELTAS; BOLD = CI EXCLUDES ZERO; OTHERWISE WITHIN RUN-TO-RUN NOISE.

	Simple Prompt			Complex Prompt		
# Distractors	π_0	**+CGVD**	**Δ [95% CI]**	π_0	**+CGVD**	**Δ [95% CI]**
0	87.0	85.0	-2.0 [-13.3,+9.3]	86.0	87.0	+1.0 [-5.8,+7.8]
1	79.0	80.0	+1.0 [-7.1,+9.1]	74.0	69.0	-5.0 [-26.9,+16.9]
2	76.0	79.0	+3.0 [-14.9,+20.9]	69.0	77.0	+8.0 [-9.9,+25.9]
3	77.0	75.0	-2.0 [-20.4,+16.4]	64.0	74.0	+10.0 [-7.6,+27.6]
4	75.0	70.0	-5.0 [-24.6,+14.6]	57.0	73.0	**+16.0 [+1.2,+30.8]**

Table II reveals a two-sided dependence on descriptive density. The baseline degrades only mildly with clutter under the simple prompt, but collapses under the compositional one, dropping from 86.0% at zero distractors to 57.0% at four, confirming the bag-of-words brittleness. For CGVD, the complex prompt unlocks the benefit: SAM 3 exploits the rich attribute phrase to isolate the correct instance, holding a stable floor under the compositional query, with the $+16.0$ pp gain at four distractors the one delta whose confidence interval excludes zero. Under the simple prompt, by contrast, CGVD provides no reliable benefit: "green spoon" gives the segmenter too little signal to separate the target from attribute-conflicting spoons. Both arms were re-executed under the instruction-consistent protocol: earlier runs gave the segmenter "spoon with green handle" under both prompt structures—scene knowledge not derivable from the simple instruction; removing the asymmetry lowered CGVD's simple-prompt numbers, a correction against our own interest. The five-seed sample matches the original protocol; per-cell values are descriptive. CGVD converts descriptive density into robustness; it does not rescue a prompt that lacks it.

D. Ablation Studies

We ablate at a single operating point—*put spoon on towel*, π_0, 18 semantic distractors (Table III)—using π_0 for its headroom; the attributions hold at this operating point only, since §IV-B shows the net effect reverses elsewhere.

Removing the Two-Layer Target Refinement reduces SR from 77.5% to 65.0%. Without cross-validation, genuine targets are erroneously inpainted out of the scene, meaning CGVD's own protection fails, which is why we describe

TABLE III

ABLATION STUDY (π_0, *put spoon on towel*, 18 SEMANTIC DISTRACTORS; 200 EPISODES/ROW). ROWS REMOVE OR SUBSTITUTE ONE COMPONENT. SR: MEAN ± SD OVER PER-SEED RATES; Δ VS. FULL PIPELINE WITH 95% CI FROM PAIRED DELTAS. BASELINE/FULL ROWS REUSE THE CORRESPONDING TABLE I RUNS; THE ROBOT-MASK EFFECT IS NOT DISTINGUISHABLE FROM ZERO.

Configuration	SR (%)	Δ [95% CI]
Baseline (no CGVD)	43.0 ± 16.0	-34.5 $[-48.2, -20.8]$
CGVD (full pipeline)	$\mathbf{77.5} \pm 6.3$	—
LaMa → mean-color fill	56.5 ± 16.0	-21.0 $[-35.3, -6.7]$
– Two-layer target refinement	65.0 ± 8.8	-12.5 $[-20.3, -4.7]$
– Robot mask protection	73.0 ± 7.1	-4.5 $[-10.4, +1.4]$

it as conditional (§III-A). Auditing a fresh 200-episode re-execution of the headline condition against the renderer's ground-truth target mask, the inpainting mask overlapped the target at $t{=}0$ in 4 of 200 episodes (2.0%), each an essentially complete erasure. The protection is therefore strong but probabilistic in practice, differing from sensitivity-probe approaches in failure rate rather than failure mode. The re-execution also replicates the headline cell (76.0% vs. 77.5%). Substituting a mean-color fill for LaMa incurs the largest single drop, to 56.5%, despite identical masks and removed content; we hypothesize, without measuring it, that the stark region boundaries act like out-of-distribution patches to the ViT backbone. Notably, a majority of the gain at this operating point thus depends on how the removed regions are filled rather than on the removal alone. Finally, removing Robot Mask Protection reduces SR to 73.0%. This -4.5 pp effect is not statistically distinguishable from zero at our sample size, so we treat robot-mask protection as qualitatively motivated (without it the compositing mask occasionally occludes the arm) rather than as a measured gain.

Deployable robot mask. To measure the cost of the renderer mask's simulation privilege, we re-ran the full pipeline with a per-frame SAM 3 prediction of the robot mask on the same seeds. No accuracy difference was detected—74.0% against 76.0% for a ground-truth-mask re-execution (paired $\Delta = -2.0$ pp)—though at this sample size the experiment only excludes accuracy costs larger than about 11 pp. The costs lie in latency and reliability: the deployable path adds 240.7 ± 1.9 ms per frame (roughly 0.96 s per chunk), and SAM 3 returns no robot detection in 20.1% of the 12,400 audited frames, falling back to the previous mask and leaving the arm's leading edge unprotected over formerly inpainted regions. Forward-kinematics projection (§III-F), which avoids this failure class entirely, remains the sensible hardware path.

Vocabulary sensitivity. Because $\mathscr{D}$ coincides with the injected inventory (§III-A), we re-ran the headline condition on the same seeds with two frozen vocabularies to measure that privilege. With a vocabulary that misses the injected clutter ($\mathscr{D} = \{$ladle, whisk, bowl, cup$\}$), success collapses to 45.5%, statistically indistinguishable from the 43.0% baseline ($\Delta = +2.5$ $[-6.6, +11.6]$). With a generic 12-category tabletop vocabulary not derived from the injection list (fork, knife, spatula, scissors, ladle, whisk, plate, bowl, cup, mug, pan, pot), success is 76.5%, statistically indistinguishable from the oracle vocabulary ($\Delta = +0.5$ $[-4.7, +5.7]$; vs. baseline $+33.5$ $[+22.3, +44.7]$). CGVD's benefit therefore requires vocabulary coverage of the clutter, but not knowledge of the exact inventory. The miss condition is not a no-op: the pipeline still runs, the scene is still edited (mask coverage falls to roughly 2% from 14%), and the erasure and blending risks remain live—matching the baseline means the benefit is forfeited, not the intervention absent. Usefully, a near-empty distractor mask is a one-line vocabulary-miss detector.

E. Comparison with a Run-Time Intervention Baseline

We compare CGVD against BYOVLA [8], the closest prior run-time observation intervention, at the headline condition (same 10 matched seeds), reimplementing its reference code with three confound-removing substitutions: GPT-4o replaced by the same closed vocabulary $\mathscr{D}$ CGVD receives, GroundingDINO+SAM2 by SAM 3, and Octo's fixed-key stochastic probe by fixed-seed π_0 forward passes. The mechanism is kept faithful (per-object Gaussian-blur probe, reference threshold 0.002 m on translation deltas, sensitive-region inpainting with the shared LaMa inpainter, per-chunk cadence as in the reference), and the probe demonstrably discriminates (mean 2.0 of 32.1 candidate regions flagged per chunk).

BYOVLA reaches 32.5%, numerically below the 43.0% baseline ($\Delta = -10.5$, within run-to-run noise) and far below CGVD's 77.5% (paired $\Delta = -45.0$, 95% CI $[-56.4, -33.6]$). Its cost is also substantial: each intervention takes 11.3 ± 2.0 s (roughly 33 policy forward passes), and per 4-step action chunk this amounts to 11.6 s against CGVD's 0.38 s, about 30×. We attribute this to per-chunk re-segmentation and re-inpainting, which makes the background flicker across chunks—precisely the fill-appearance instability the mean-color-fill row of Table III exposes, and that CGVD's cached clean plate avoids—and to the conservative threshold leaving most of the 18 distractors in view. Two caveats apply: this is our reimplementation, and eighteen dense distractors lie well outside BYOVLA's original regime of a real robot with a handful of distractors; we also did not tune its threshold. The fair conclusion is that per-chunk probe-and-inpaint does not scale to dense clutter, not that the method fails in its regime.

F. Latency Analysis

CGVD concentrates its expensive operations on the initialization frame; afterwards the system only composites against the cached clean scene. Timings were measured on an NVIDIA A10G across the evaluation runs (means ± SD over episodes); an earlier version reported a single blended per-step figure, replaced here by decomposed values. The policy forward pass is unaffected (318.1 ± 6.6 ms over 500 baseline episodes vs. 318.5 ± 5.5 ms over 1,100 with CGVD; one forward per 4-step chunk), and per-frame compositing adds 15.6 ± 0.4 ms—62 ms per chunk, roughly 10% additional episode wall time excluding initialization ($9.11 \rightarrow 10.00$ s in the attribute runs). Initialization is a one-off per-episode cost growing with prompt count, from 0.68 ± 0.29 s in the attribute scenes to 0.91 ± 0.02 s at 18 distractors (measured over warm episodes; a session's first episode additionally pays model loading). The overhead is modest for quasi-static tabletop tasks, less so for time-critical settings.

V. Limitations

CGVD's validity envelope is defined by the following assumptions.

Static scene. With the clean scene cached at $t{=}0$ and α fixed (§III-F), a moved distractor desynchronizes the cache, and an object entering later can be blended toward the cached background while physically present—outside access-controlled workspaces a hazard, not merely an accuracy loss. Continuous re-segmentation would lift the assumption but is prohibitive at control rate.

Contextually informative clutter. On *put carrot on plate* CGVD never outperforms the baseline and significantly underperforms it at several densities (up to -14.5 pp for GR00T; §IV-B).

Simulation-only evaluation. No real-robot experiment is reported, and two inputs are simulation-privileged: the renderer robot mask and the per-episode inventory $\mathscr{D}$. SAM 3 recall over $\mathscr{D}$ and LaMa fill fidelity on real textured scenes (the largest single factor in Table III) are unvalidated on hardware, and FK projection is untested.

Single target instance. Layer 2 keeps only the top-scoring component (§III-C); a scene with two genuine target instances would lose one.

Occlusion fabrication and in-flight blending. Content the arm occluded at $t{=}0$ is synthesized, not observed, in the cached scene (Eq. 6); and since α protects the target only at its $t{=}0$ location, the moving target crosses inpainted regions—by over 10% of its ground-truth pixels at some frame in 35% of the 200 audited headline episodes—exposing the grasped object to partial blending.

VI. Conclusion

We introduced CGVD, a training-free inference framework that mitigates the Precision-Reasoning Gap through language-grounded segmentation and inpainting. Across more than 20,000 SimplerEnv episodes with π_0 and GR00T, CGVD prevents performance collapse under dense semantic clutter and improves attribute adherence in an exploratory compositional-prompt condition, relative to the unaugmented policies; at the headline condition it outperforms a vocabulary-matched BYOVLA reimplementation by 45 pp at a thirtieth of its per-chunk cost (in a regime outside its design envelope, threshold untuned). The benefit is conditional: where contextual clutter aids the policy, CGVD never outperforms and often significantly underperforms the baseline. Bounded by static-background and closed-vocabulary assumptions, it emerges as a practical defense against semantically confusable clutter. Future work will explore real-time mask updating, intrusion detection, open-vocabulary distractor sets, and criteria for when to distill.

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